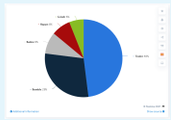
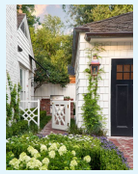
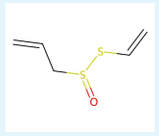
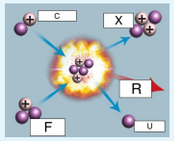
Model-Agnostic Seed & Rubric Collection

Data Pool



	Red	Green
Big	2	1
Small	5	3



Seed Mining

Qwen2.5-VL

TextHawk2

InternVL3.5

Qwen3-VL

Rubric Drafting

Generation + Aggregation

Weight	Criterion	Reference
●		
●		
●		
●		

Instruction-Rubric Pool



Essential

Additional

Essential

Additional



On-Policy Data Curation & Optimization

Sampling

Base Model



Rollout Engine

y_1
y_2
y_3
.....
y_{32}

Scoring

Essential

Additional

VLM-as-a-Judge

y_1	r_1
y_2	r_2
y_3	r_3
y_4	r_4
y_5	r_5
y_6	r_6
.....	
y_{32}	r_{32}

Pair Mining

Qualification Filtered

Margin

.....

Rules

· Qualification
· Margin
· Learnability
· Diversity

✓
✗

.....

✓
✗

Iterative DPO

Initial Reference + Split 1

Intermediate Reference + Split 2

Final Reference + Split 3